

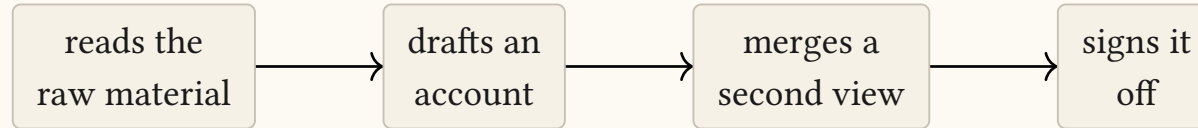


Keeping meaning between machines

Why a document format, and what it is measured to save

A pipeline is a chain of handoffs

Something reads. A model writes. A second model merges. A person signs off. Each step passes its result to the next.



Nothing here is new, and none of it needs AI. Models only made the handoffs cheap enough to do many times.

What crosses each arrow is prose

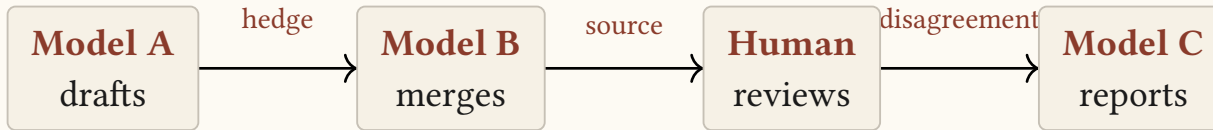
It is the only thing every participant can both write and read. So it is what gets used.

But the model that wrote it knew more than it said. It knew which number came off a dashboard and which sentence was its own guess. It knew the canary data pointed the other way.

Then it wrote paragraphs, because paragraphs were the only thing on offer.

The structure existed. It just had nowhere to go.

Three things go missing, every hop



Hedges. “suggests”
becomes “shows” becomes
“we found.”

Sources. A citation does
not survive a paraphrase.

Disagreement. Two
views go in, the winner
comes out.

Prose never tells you what it dropped

A summary that lost every citation looks exactly like a summary that had none to lose.

No error. No warning. No diff.

Every step reports success, and the artifact quietly gets worse. This is the whole problem.

So we measured it

Five documents. Five hops each. Two models. Ninety claims. A different model reads the far end and reports what survived.

	tokens	claims kept	hedges lost	sources kept
no summarising	1.00	90/90	0/90	50/50
prose, five hops	0.29	72/90	42/90	1/50
smysl, five hops	0.49	90/90	0/90	50/50

Of fifty claims that named where they came from, **one** still named it. Not one in ten. One.

The top row is why you can believe the other two

Same judge. Same prompt. Reading each document **before** any summarising happened.

It found every hedge and every source, and declined to rule on nothing. If the judge were blind, that row would show losses too.

So what went missing went missing **in the chain** — not in the instrument measuring the chain.

“Just use a schema”

Pass JSON between the steps instead of text. This genuinely fixes the machine-to-machine problem — a `confidence` field does not evaporate.

Then the artifact becomes `{"claims":[{"id":"c-0417","conf":0.6}]}` and the person in the chain stops reading it.

Nobody reviews a wire format. The one place a human could have caught the drift is now the one place they cannot see.

“Just tell the model to be careful”

Put **preserve all hedges and citations** in the prompt. Worth doing. It helps, somewhat.

Now: when the summariser returns, what checks whether it complied?

Nothing does. There is no assertion to fail, no exit code, no diff.

You have added an intention to a system that cannot notice intentions being broken.

One artifact, for the machine and the person

```
@evidence e/pool-wait { status: measured, source: { kind: metric, ref: "pool.wait_ms" } }  
~ Pool acquisition wait rose from 2 ms to 310 ms.  
  
@claim c/pool-saturation { status: inferred, grounds: [e/pool-wait] }  
~ The eu-west connection pool is saturated.  
  
@rel c/canary-clean --rebutts--> c/pool-saturation { weight: 0.6 }
```

There is no separate machine version. The bytes that cross the wire are the bytes a reviewer opens.

Confidence can fall. It can never rise.

measured — an instrument recorded it

cited — a source you can open says so

derived — computed from what is above

inferred — a model reasoned it out

speculative — offered, not yet grounded

a hop may
only move
down

Every claim carries how sure the document is entitled to be.

It is a field a program checks — not a tone of voice a summariser can talk itself out of.

Two rules stop a guess becoming a fact

Rule T — at the door

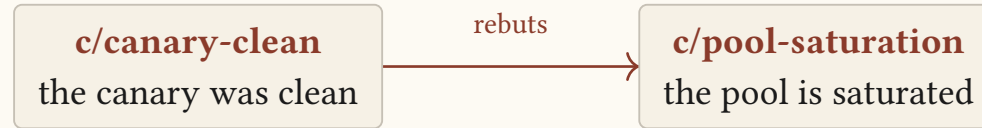
A model reasoning from its own knowledge may claim *inferred* and no more, however confidently it writes.

Rule M — inside the graph

A claim can never be stronger than the weakest thing it rests on.

Checked mechanically, on every claim, every time — not left to whoever is doing the summarising.

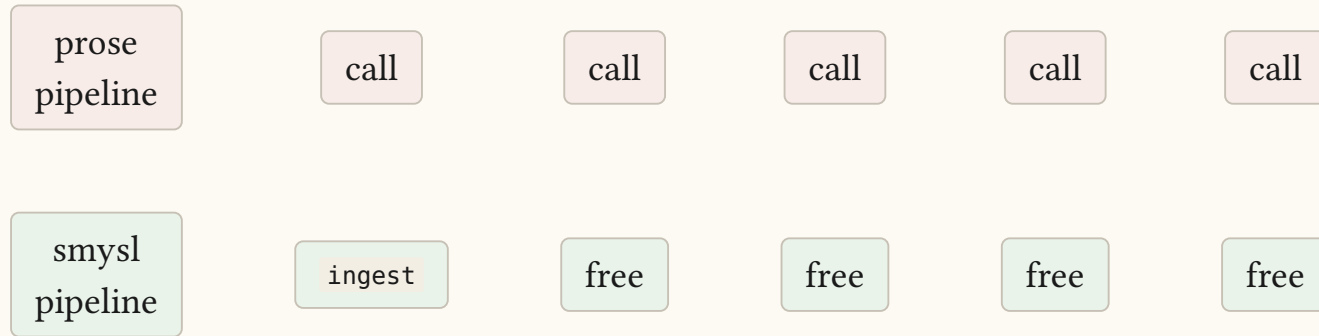
Disagreement is part of the document



The objection sits next to the claim it argues with — not in a review thread nobody opens twice.

Cutting a document to fit a budget must keep the rebuttal with the claim. A one-sided summary is not an option the tool offers.

Pay a model once, at the entrance



Everything after the entrance — merging, selecting, ordering, rendering, checking — is ordinary computation. Same input, same bytes out, forever.

Five model calls on the prose side of that measurement. Zero on the other.



A claim and its citation travel as one thing

from the model that wrote it to the person who checks it.

Nothing is translated, so nothing is lost in translation.

github.com/vulogov/smysl

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